

OFDM $\rightarrow 64$

40M Hz, $\frac{5}{6}$, 800ns

QAM 64

$$\begin{aligned} 40\text{M Hz} &\Rightarrow 40\text{M samples} \text{ --- } 1\text{Sec} \\ 64 \text{ time samples} &\approx \frac{64}{40\text{M}} \\ &= 1.6 \mu\text{S} \end{aligned}$$

Guard band = $0.8 \mu\text{S}$

$$\begin{aligned} \text{OFDM symbol duration} &= 1.6 + 0.8 \\ &= 2.4 \mu\text{S} \end{aligned}$$

QAM 64 $\rightarrow 6 \text{ bits} \times \frac{5}{6}$

5x48 bits --- $2.4 \mu\text{S}$

$$\begin{aligned} \Rightarrow \frac{2.4 \mu\text{S}}{10^6} &\text{ --- } 5 \times 48 \\ &\rightarrow \frac{5 \times 48}{2.4} = 100 \text{ Mbps} \end{aligned}$$

No, OFDM

1 Sub-carrier only

40M time samples $\rightarrow$ 1 sec

1 time

$$\frac{1}{40M}$$

$$= 0.025 \mu s$$

$$\begin{aligned} 1 \text{ OFDM Symbol} &= 0.025 \times 0.8 \\ &= 0.02 \mu s \end{aligned}$$

$$QAM 64 \rightarrow \frac{6 \times 5}{6}$$

$$5 \text{ bits} \rightarrow 0.02 \mu s$$

$$\Rightarrow 10^6 \mu s \rightarrow \frac{5}{0.02} \times 10^6$$

$$\approx 6 \text{ Mbps}$$

